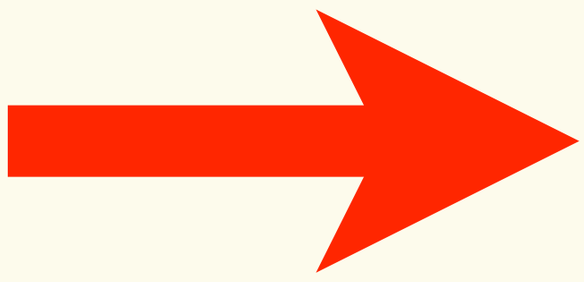
Python for LLM & Gen Ai 14-Day Foundation Plan



First Two weeks (**Days 1–14**) Python-for-LLM/GenAI study showing how all subtopics connect logically from **Core Syntax** → **Applied GenAI foundations**.





Python for LLM & Gen Ai
14-Day Foundation Plan



Week 1 : Core Python Foundations for AI (Days 1–7)

Day 1 – Python Setup & Environment
- Install Python, pip, venv
- Jupyter / VS Code setup
- Task: Create “Hello LLM” notebook

Day 2 – Syntax, Variables, Data Types
- Strings, Lists, Tuples, Dicts
- Type casting, mutability
- Task: Mini quiz + type manipulations

Day 3 – Control Flow & Loops
- **if-else**, **for**, **while**
- List comprehensions
- Task: Implement token counter loop

Day 4 – Functions & Scoping
- Defining functions, arguments, **return** types
- Lambda, ***args**, ****kwargs**
- Task: Create reusable “prompt_cleaner()”

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Week 1 : Core Python Foundations for AI (Days 1–7)

Day 5 – Object-Oriented Python (OOP)
- ├── Classes, objects, inheritance, composition
- ├── Use-case: “PromptTemplate” class
- └── Task: Build class managing prompt templates

Day 6 – File Handling & I/O
- ├── Reading/writing .txt, .csv, .json
- ├── Working with large corpus files
- └── Task: Script to read text corpus & store tokens

Day 7 – Exception Handling & Debugging
- ├── **Try-except,** raising custom errors
- ├── Logging errors for LLM pipelines
- └── Task: Add logging to file reader + prompt cleaner

Week 1 Project:
- ├── Integrate OOP + loops + I/O + exceptions
- ├── Prompt Data Manager reads corpus, cleans, & logs
- └── Deliverable: prompt_data_manager.py

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Week 1 : Intermediate Python for GenAI (Days 8–14)

Day 8 – Python Data Structures for AI
- ├── Stacks, queues, dictionaries for memory simulation
- ├── Generators & iterators
- └── Task: Build conversation memory store

Day 9 – Functional & Modular Programming
- ├── Map, filter, reduce
- ├── Decorators, closures **for** pipeline functions
- └── Task: Decorator **for** timing model calls

Day 10 – Working with External Libraries
- ├── **Installing & importing**: NumPy,Pandas,Requests
- ├── Using APIs with requests (simulate model query)
- └── Task: Fetch mock LLM response from API endpoint

Day 11 – Data Serialization
- ├── JSON, Pickle, CSV I/O
- ├── Saving conversation states
- └── Task: Serialize chat sessions as JSON

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Week 1 : Intermediate Python for GenAI (Days 8–14)

Day 12 – Virtual Environments & Package Mgmt
- pip, conda, Poetry basics
- requirements.txt / pyproject.toml
- Task: Package “Prompt DM” as installable module

Day 13 – Unit Testing & Code Quality
- pytest, unittest
- Linting with flake8 / black
- Task: Write tests for your data manager functions

Day 14 – Mini Capstone: Clean Pipeline
- Combine all (data ingestion → processing → output)
- Logging + error handling + modular design
- Deliverable: `text_preprocessing_pipeline.py`

Week 2 Outcome:
- Confident with Python fundamentals for AI
- Familiar with file I/O, modularity, testing, & API integration